

Roster IQ: A Benchmark Player Clustering Approach to Evaluating Player Success, Fit, and Value in DI Men's College Basketball

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Motivation

Traditional analytics seeks to capture overall player performance, lacking the necessary *nuance* of a player's role in a team game.

- Roster IQ** was created to pioneer a *new* methodology of roster construction analytics that emphasizes **interpretability and role-awareness**
- Our methodology compares player candidates against an artificial **benchmark player** to evaluate success, compatibility, and differential value relative to a known, data-driven standard

Methods

Data Collection

Collected a 35,000+ player and team SQLite database from Bart Torvik, University websites, and ESPN. Normalized statistics to per 100 possessions

Preprocessing

Applied principal component analysis and k-means clustering on teams and players – maximizing cluster heterogeneity index, silhouette score, and explained variance

Metric Calculation

Benchmark Player (BP): historical average prototype player at a specific role

Fit Score (FS): *cosine similarity* with BP on style metrics. Closer to +1 represents a better player role "fit"

Value Over Clustered Benchmark Player (VOCBP): *z-score difference* from BP on production metrics. A higher VOCBP reflects greater player value from BP

Role Value Index: *weighted z-score sum* of FS and VOCBP (60/40 split)

Benchmark Success Score: end-of-season *z-score differential* from BP on impact metrics. Successful season: >-0.05

Results

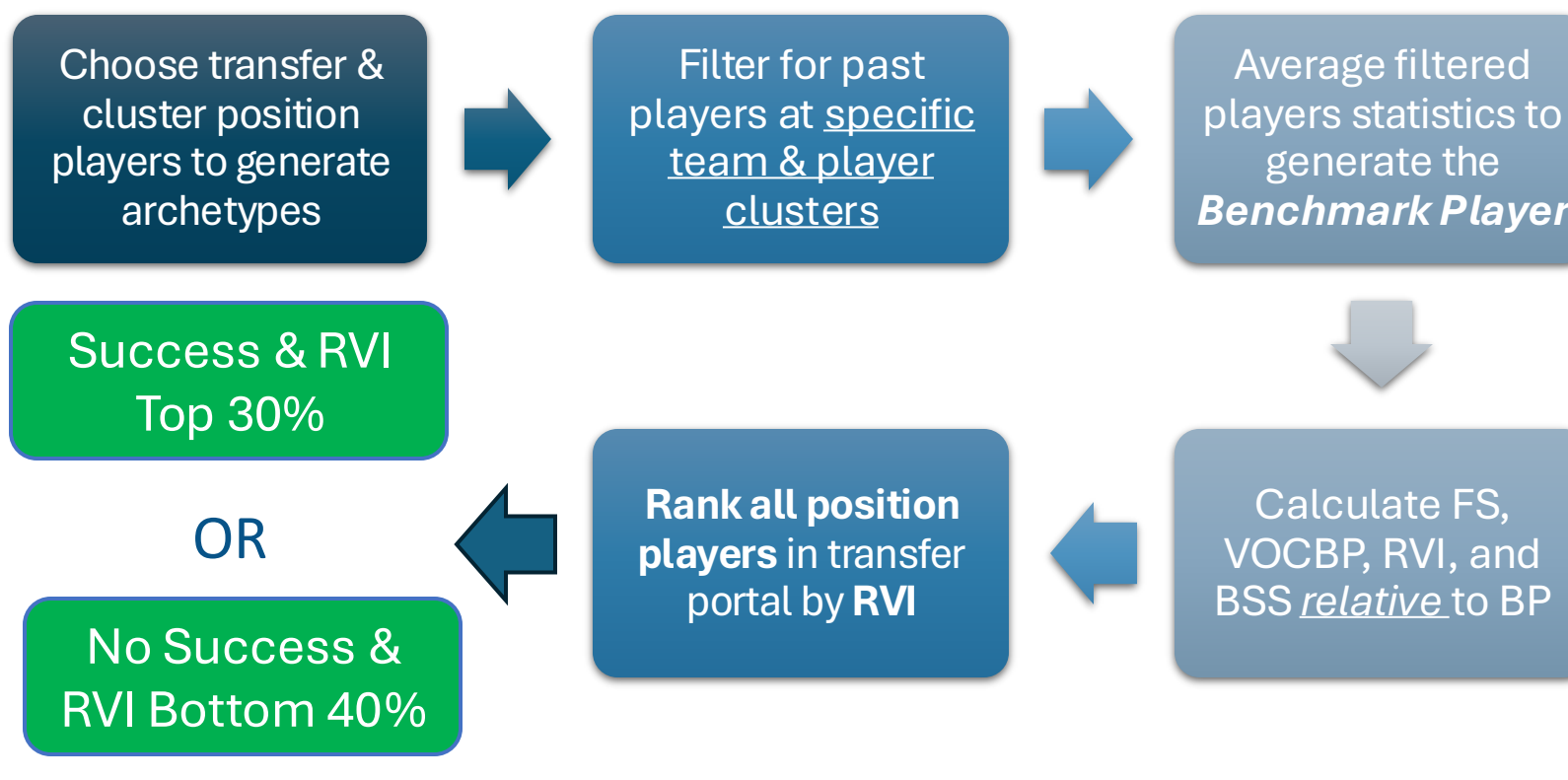
Historical Testing Results

Condition	Total	Correct	Accuracy	Chance	Δ	Success%	Unsuccessful%	p(H1: Acc>p0)
ESS >= 30 (run 1)	150	80	53.3%	35.4%	17.9%	46.3%	53.7%	5.35e-06
ESS >= 30 (run 2)	150	80	53.3%	35.3%	18.0%	47.5%	52.5%	4.78e-06
ESS >= 30 (run 3)	454	239	52.6%	35.4%	17.2%	45.6%	54.4%	9.31e-14
Top 25% Barthag + ESS >= 30	466	242	51.9%	33.1%	18.8%	69.4%	30.6%	2.04e-16
ESS < 30	195	103	52.8%	36.4%	16.4%	35.9%	64.1%	2.18e-06
No ESS threshold	647	335	51.8%	35.6%	16.2%	44.2%	55.8%	8.97e-17

Team PCA & Cluster Overview

Season	Num Clusters	Mean Cluster Heterogeneity Index (CHI)	Silhouette Score	Explained Variance (%)
2021	10	0.302	0.224	64.0
2022	9	0.294	0.228	63.6
2023	9	0.286	0.224	64.1
2024	9	0.288	0.214	63.9

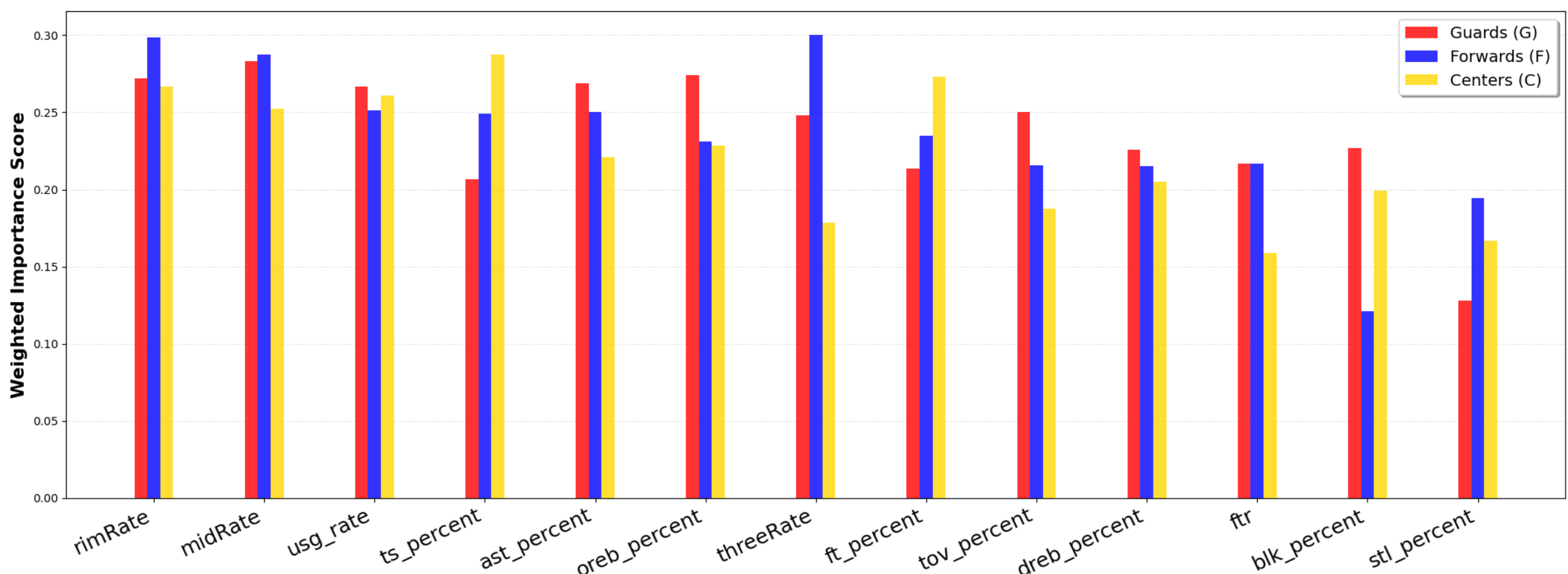
Historical Testing Process



Player PCA & Cluster Overview

	2021			2022			2023			2024		
	G	F	C	G	F	C	G	F	C	G	F	C
Num Clusters	13	13	9	12	13	8	12	13	8	11	13	8
Mean CHI	0.342	0.321	0.380	0.322	0.325	0.365	0.313	0.320	0.368	0.302	0.332	0.366
Silhouette Score	0.156	0.151	0.159	0.155	0.152	0.165	0.157	0.163	0.161	0.149	0.146	0.158
Explained Var (%)	62.3	60.7	54.8	61.7	59.4	55.2	62.0	59.3	55.6	61.8	59.1	55.2

Weighted Absolute PCA Importance of Player Statistics by Position



Pelle Larsson – 2021

Univ of Arizona Alum



	FS	VOCBP	RVI	SOS Adj	RVI Rank	ESS	BSS
raw	0.53	0.68	-				
z-score	-0.32	1.63	0.46				
t-score	-	-	59.78	0.149	63/408	39.02 > 30	0.0382 > -0.05
Archetype Split							
Player (Dual)				Team (Single)			
Defensive Stopper		Slasher		Elite Rebounders & Paint Beasts			
60.590%		39.410%		100.000%			

Next Steps

I am now seeking feedback from basketball professionals on the methods and **aim to apply the tool *prospectively* with coaches**. I hope the "benchmark player" approach can inspire a new wave of sports player analytics.

Potential improvements with additional time and resources:

- Cluster players by roles, not positions, as the game grows more "positionless"
- Tune k-means clustering model and/or test other clustering algorithms
- Build a regression model within clusters to predict other analytics
- Adjust stats by schedule, location, etc.
- Gather advanced basketball data like shot location, drive rates, etc.

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